

AI-Powered Debt Relief & Financial Recovery Platform

Project Report

1. Introduction

1.1 Project Overview

The AI-Powered Debt Relief & Financial Recovery Platform is a full-stack web application that helps borrowers who have fallen behind on loan payments understand how serious their situation is and negotiate a fair settlement with their lender. It combines a rule-based financial-analysis engine with generative AI to turn raw loan numbers into an actionable, personalized negotiation letter.

1.2 Purpose

Borrowers who fall behind on EMIs often don't have an easy way to quantify how serious their situation is, or the confidence and vocabulary to negotiate with a lender. This platform closes that gap: it scores financial stress objectively from the borrower's own numbers, recommends a realistic settlement range, and drafts a lender-ready negotiation letter in the borrower's chosen tone — reducing the anxiety and guesswork around debt recovery.

2. Ideation Phase

2.1 Problem Statement

Borrowers who have missed loan payments struggle to (a) objectively gauge how severe their financial stress is, and (b) communicate with their lender in a way that is confident, factual, and likely to result in a fair settlement — leading to avoidable defaults, damaged credit, and prolonged financial hardship.

2.2 Empathy Map Canvas

Built around a borrower persona who has recently missed one or more EMI payments:

- Says: "I don't know how bad this actually is" / "I don't know what to say to my lender"
- Thinks: worries the lender will be inflexible or that the loan will be reported as default
- Does: avoids lender calls, searches online for generic debt advice, delays taking action
- Feels: anxious, overwhelmed, and unsure whether asking for a settlement is even reasonable

2.3 Brainstorming

Ideas generated and grouped into themes (assessment, communication, tracking), then prioritized by impact, feasibility, and effort:

Idea	Impact	Feasibility	Effort
Rule-based debt-stress score & settlement recommendation	High	High	Low
AI-generated, lender-specific negotiation letters	High	Medium	Medium
Financial health dashboard with settlement history	Medium	High	Low

Multi-loan support per borrower	Medium	High	Low
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3. Requirement Analysis

3.1 Customer Journey Map

A 5-stage borrower journey — Awareness → Loan Intake → Financial Analysis → Negotiation → Resolution — mapping the borrower's steps, touchpoints, goals, positive/negative moments, and opportunities at each stage (see the Borrower Journey Map deliverable).

3.2 Solution Requirement

12 functional and 7 non-functional requirements were defined, along with assumptions and constraints:

Type	Requirement
FR-1	Borrower can create a loan record with outstanding amount, EMI, overdue months, and monthly income
FR-2	System computes EMI ratio and monthly surplus from loan data
FR-3	System computes a debt-stress score (0–100) and maps it to Low/Moderate/High/Critical
FR-4	System recommends a settlement percentage, clamped between 35% and 75%
FR-5	Borrower can select a negotiation letter type and tone
FR-6	System generates a personalized letter via Gemini, grounded in the borrower's real numbers
FR-7	System falls back to a template-based letter if no Gemini API key is configured
FR-8	Borrower can view past negotiation letters for a loan
FR-9	Dashboard aggregates financial health stats across all of a borrower's loans
FR-10	Borrower can delete a loan
FR-11	System exposes a health-check endpoint for monitoring
FR-12	Frontend and backend communicate over a documented REST API
NFR-1	Loan analysis responds in under 1 second (no external calls)
NFR-2	Gemini API key is never committed to source control
NFR-3	Backend and frontend are deployed and hosted independently (Render / Vercel)
NFR-4	System remains usable (via template fallback) if the Gemini API is unavailable
NFR-5	CORS restricts backend access to the deployed frontend origin
NFR-6	Codebase is version-controlled with a clear router/service separation
NFR-7	UI follows a consistent design system (ink-ledger) across all pages

3.3 Data Flow Diagram

A Level 0 context diagram (borrower ↔ platform ↔ Gemini API) and a Level 1 process breakdown (loan intake → financial analysis → letter generation → dashboard aggregation) were produced, backing 7 user stories with acceptance criteria.

3.4 Technology Stack

Layer	Technology
Frontend	React 18 (Vite), React Router v6, custom CSS
Backend	FastAPI (Python), Pydantic, Uvicorn
ORM / Database	SQLAlchemy + SQLite
AI	Google Gemini API (gemini-1.5-flash), with a template-based fallback
Backend hosting	Render
Frontend hosting	Vercel
Version control	Git + GitHub

4. Project Design

4.1 Problem Solution Fit

The core insight is that borrowers don't need a loan officer — they need an objective number (the debt-stress score) and a well-worded first draft (the negotiation letter). Automating both removes the two biggest blockers to a borrower taking action: uncertainty about severity, and uncertainty about how to ask.

4.2 Proposed Solution

Three integrated scenarios: (1) AI-powered settlement recommendation from loan inputs, (2) Gemini-generated negotiation letters grounded in those same numbers, and (3) a dashboard that tracks financial health and settlement history over time, so the borrower sees whether their situation is improving.

4.3 Solution Architecture

See the Technology Stack (Architecture & Stack) deliverable for the full architecture diagram, component table, and application-characteristics table.

5. Project Planning & Scheduling

5.1 Project Planning

Work was broken down by scenario (settlement recommendation, letter generation, dashboard) and by layer (backend routers/services, frontend pages/components), with the financial-analysis engine and Gemini integration built as independently testable services. [Insert sprint/milestone breakdown and story points from the Project Planning Phase document here.]

6. Functional and Performance Testing

6.1 Performance Testing

Functional tests cover loan input validation, debt-stress/settlement calculation accuracy, negotiation letter generation, and the Gemini API connection with fallback. Performance tests cover response time for analysis and letter generation, concurrent request handling, and dashboard load with a large number of loans (see the Functional & Performance Testing and UAT deliverables for full test cases and results).

7. Results

7.1 Output Screenshots

[Insert screenshots of the Dashboard, Add Loan form, Loan Detail / debt-stress gauge, and negotiation letter generation screens here.]

8. Advantages & Disadvantages

Advantages:

- Gives borrowers an objective, formula-based read on their financial stress instead of guesswork
- Produces a lender-ready negotiation letter in seconds, using the borrower's real numbers
- Keeps working even without an AI API key, via the template-based fallback
- Tracks settlement history so borrowers can see progress over time

Disadvantages:

- Debt-stress scoring is rule-based, not personalized by lender-specific policies or regional regulations
- No authentication in the current scope, so the platform is not yet suitable for multi-user production use
- Settlement outcomes still depend entirely on the lender's willingness to negotiate

8.1 Known Issues / Anticipated Risks

Reliability & external dependency:

- Gemini API latency or rate limits could slow or fail letter generation; the template fallback needs to trigger on timeouts, not just on a missing API key
- Cold starts on Render's free tier can make the first request after idle time noticeably slow

Data & concurrency:

- SQLite handles concurrent writes poorly — simultaneous requests to /loans or /negotiation/generate may hit "database is locked" errors under real load
- No pagination on GET /loans or GET /dashboard/summary — response time and payload size will degrade as a borrower's loan count grows

Correctness edge cases:

- Zero or negative monthly income / EMI / outstanding amount can break the EMI ratio and stress-score math unless explicitly guarded in financial_analysis.py
- Very short overdue periods combined with very high EMI ratios can push the settlement percentage to its 35–75% clamp in ways that feel arbitrary without an explanation shown in the UI

Security / access control:

- No authentication means any client can read, edit, or delete any loan by guessing/incrementing its ID — acceptable for a demo, not for real borrower data
- CORS should be confirmed as restricted to the deployed frontend origin and not left open from local development testing

UX:

- The StressGauge SVG may not scale cleanly on small mobile viewports
- A "Critical" stress score with no immediate next-step link could increase borrower anxiety rather than help (flagged as an opportunity in the Borrower Journey Map)

9. Conclusion

The platform demonstrates that combining a transparent, rule-based financial model with generative AI can turn a stressful, ambiguous situation — falling behind on a loan — into a clear, actionable path forward for the borrower, without requiring them to understand the underlying finance or negotiation theory themselves.

10. Future Scope

- Borrower authentication and multi-user accounts
- Lender-response tracking with suggested next actions after a rejected offer
- Side-by-side comparison of multiple negotiation letter tone variants before sending
- Migration from SQLite to a managed cloud database as usage scales
- Regional/lender-specific negotiation policy tuning for the settlement recommendation

11. Appendix

Source Code: <https://github.com/bhavyainturi9/debt-relief-platform>

Dataset Link: Not applicable — the platform computes results from user-entered loan data rather than a static dataset

GitHub & Project Demo Link: <https://github.com/bhavyainturi9/debt-relief-platform>